DO NOW — RECALL

Lesson 119 · Spiral Development and Method Selection

Name: _____ Class: _____ Date: _____

1. What is one distinguishing XP practice?

2. An OR gate has inputs 0 and 1. What is the output?

3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

4. What is the time complexity of sequential search in the worst case?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

For example TDD, pair programming, CI or frequent refactoring.

2. Model answer

1.

3. Model answer

X is removed; Y remains at the front.

4. Model answer

O(n).

By the end, you can…

• Set objectives, options and constraints for a spiral iteration.

• Evaluate and reduce technical or organisational uncertainties before commitment.

• Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

Vocabulary — know these terms

spiral model — Risk-driven iterative development lifecycle.

risk analysis — Estimating likelihood and impact of potential problems.

prototype — Early model built to explore or test an idea.

iteration — Repeated cycle of development and evaluation.

NEW LEARNING

Read, annotate and explain

Plan the cycle

Set objectives, options and constraints for a spiral iteration.

Analyse risks

Evaluate and reduce technical or organisational uncertainties before commitment.

Prototype and review

Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

Model answer

Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the project.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe three central activities within one spiral development cycle. [6]

Question 2

Give two reasons a fixed-requirement small project might not benefit from the spiral model. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare waterfall, agile/RAD, XP and spiral for a high-risk changing specification. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Spiral development is simply waterfall repeated without evaluating project risks.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.